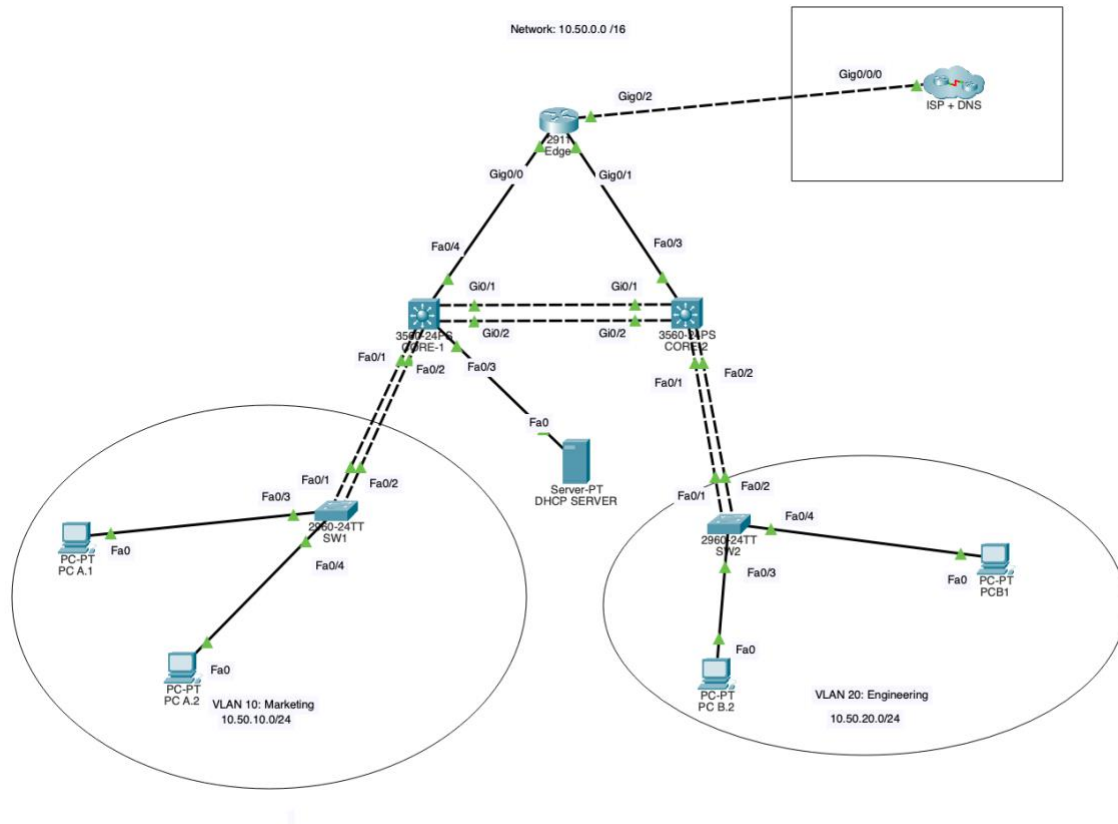


High Availability Enterprise Campus LAN

Markel Crumpler

Topology



Objective

Design a highly available campus network utilizing Layer 2 and Layer 3 redundancy to eliminate single points of failure at the core and distribution layers.

Technologies Deployed

- VLANs and 802.1Q Trunking.
- Layer 2 EtherChannel utilizing LACP.
- Rapid PVST+ Spanning Tree Protocol.
- First Hop Redundancy Protocols (FHRP) using HSRP.
- Single-Area OSPFv2 for internal routing.
- Centralized DHCP with IP Helper Addresses.

Configuration Insight

- To prevent network loops while maximizing bandwidth, I configured LACP between the Core and Access switches.
- HSRP on CORE-1 and CORE-2 to provide a virtual default gateway for VLANs 10 and 20.
- Established a virtual IP as the default gateway so that if Core-1 suffers a hardware failure, Core-2 will immediately resume routing duties without dropping client connections.

Security and Access

- Mitigated VLAN hopping attacks by explicitly disabling Dynamic Trunking Protocol (DTP) using the switchport nonegotiate command and hardcoding all 802.1Q trunks
- Isolated network infrastructure traffic to a dedicated Management/Native network (VLAN 99) and enforced SSH-only remote access on VTY lines to prevent cleartext credential interception
- Secured the enterprise boundary by designing and deploying Extended Access Control Lists (ACLs) to filter TCP/IP packets
 - Enforced strict corporate access policies by explicitly permitting the Marketing department (VLAN 10) to reach external internet targets while actively denying internet access to restricted subnets like Engineering (VLAN 20)

Verifications

ISP

ISP + DNS

Physical Config CLI Attributes

IOS Command Line Interface

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

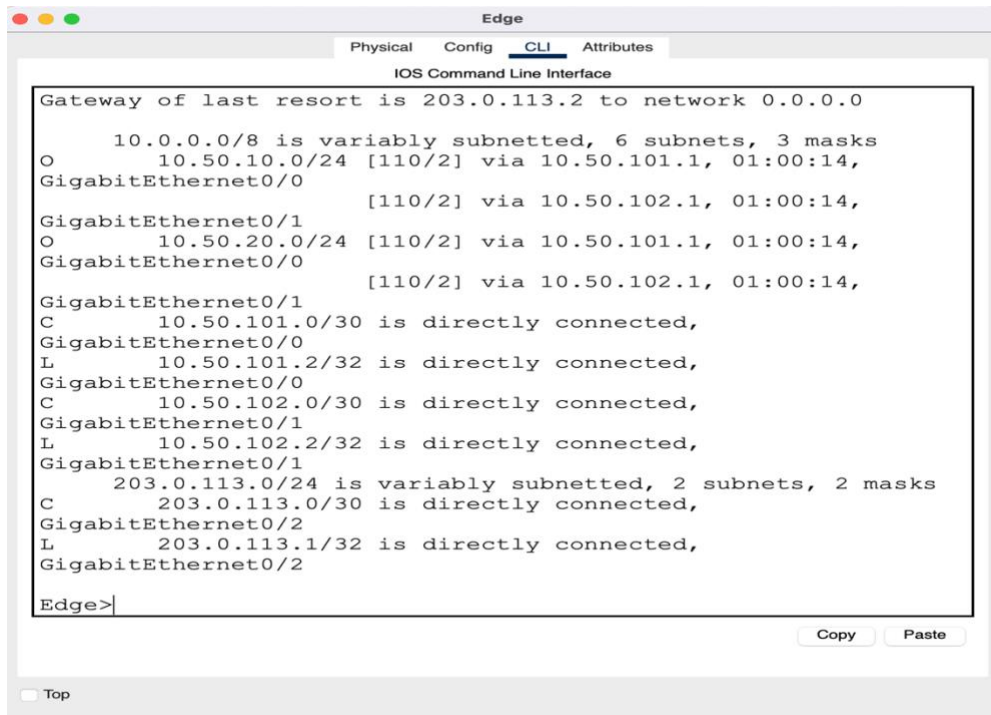
8.0.0.0/32 is subnetted, 2 subnets
C 8.8.4.4/32 is directly connected, Loopback1
C 8.8.8.8/32 is directly connected, Loopback0
203.0.113.0/24 is variably subnetted, 2 subnets, 2 masks
C 203.0.113.0/30 is directly connected, GigabitEthernet0/0/0
L 203.0.113.2/32 is directly connected, GigabitEthernet0/0/0

ISP>

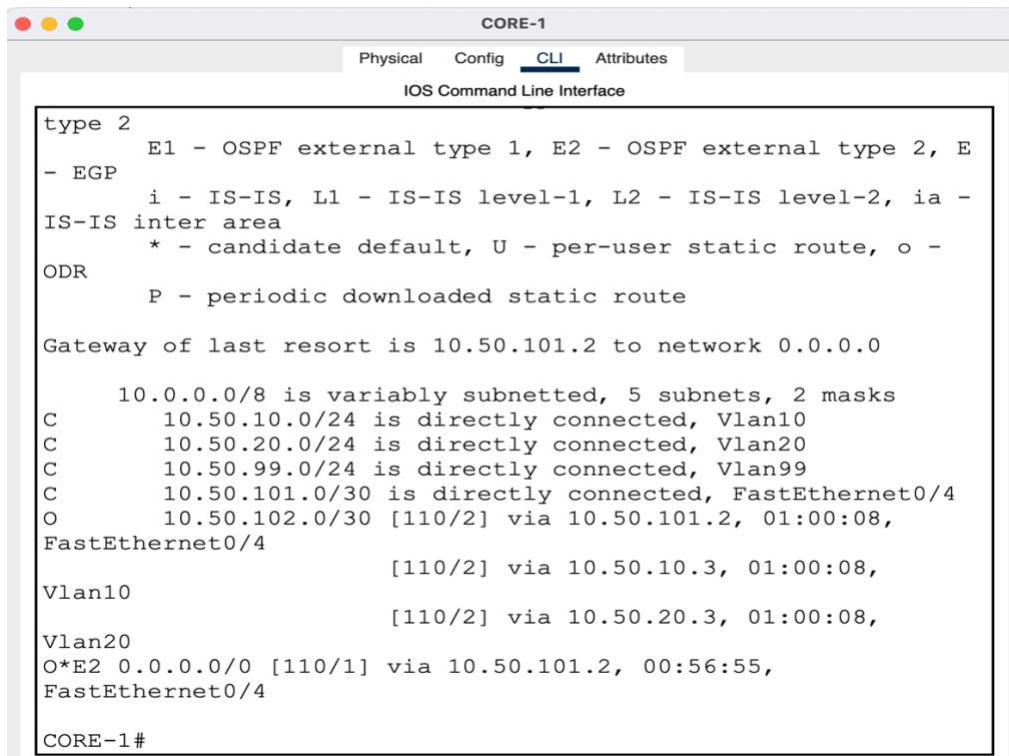
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Edge Router



Core Layer 3 Switch 1



Distribution Switch

SW2

Physical Config **CLI** Attributes

IOS Command Line Interface

```
Fa0/14, Fa0/15, Fa0/16
Fa0/18, Fa0/19, Fa0/20
Fa0/22, Fa0/23, Fa0/24

10 Marketing active
20 Engineering active
99 Management&Native active
1002 fddi-default active
1003 token-ring-default active
1004 fddinet-default active
1005 trnet-default active

Fa0/17,
Fa0/21,
Gig0/1, Gig0/2
Fa0/3, Fa0/4
```

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp
BrdgMode	Trans1	Trans2					
1	enet	100001	1500	-	-	-	-
0	0						
10	enet	100010	1500	-	-	-	-
0	0						
20	enet	100020	1500	-	-	-	-
0	0						

SW2 #

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End Device: PC B.2 ping to DNS Server IP

PC B.2

Physical Config **Desktop** Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 8.8.4.4

Pinging 8.8.4.4 with 32 bytes of data:

Reply from 8.8.4.4: bytes=32 time=31ms TTL=253
Reply from 8.8.4.4: bytes=32 time<1ms TTL=253
Reply from 8.8.4.4: bytes=32 time<1ms TTL=253
Reply from 8.8.4.4: bytes=32 time<1ms TTL=253

Ping statistics for 8.8.4.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 31ms, Average = 7ms

C:\>
```

End Device: PC A.2 using DHCP to obtain IP Address + DNS Server Address

PC A.2

PhysicalConfigDesktopProgrammingAttributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

Bluetooth

Global Settings

Display NamePC A.2

InterfacesFastEthernet0

Gateway/DNS IPv4

DHCP

Static

Default Gateway10.50.10.1

DNS Server8.8.4.4

Gateway/DNS IPv6

Automatic

Static

Default Gateway

DNS Server